Addition and subtraction of radicals

1 Simplify the following:

a $8\sqrt{3}+18\sqrt{3}$

b $-12\sqrt{5}+6\sqrt{5}-12\sqrt{5}$

c $8\sqrt{6}+17\sqrt{13}+19\sqrt{6}-6\sqrt{13}$

d $17\sqrt{6}-8\sqrt{11}+3\sqrt{6}+13\sqrt{11}$

e $\sqrt{6}+\sqrt{54}$

f $\sqrt{180}+\sqrt{500}$

g $\sqrt{32}-\sqrt{98}$

h $4\sqrt{45}+5\sqrt{180}$

i $\sqrt{10}+10\sqrt{10}$

j $-19\sqrt{5}+4\sqrt{5}$

k $18\sqrt{2}-10\sqrt{2}+20\sqrt{2}$

l $19\sqrt{3}+19\sqrt{7}-7\sqrt{7}+22\sqrt{3}$

m $\sqrt{243}+\sqrt{3}$

n $2\sqrt{75}-3\sqrt{108}$

o $\dfrac{12+9\sqrt{5}}{3}$

2 Use the formula $\sqrt{a}+\sqrt{b}=\sqrt{a+b+2\sqrt{ab}}$ to simplify the following:.

a $\sqrt{3}+\sqrt{12}$

b $\sqrt{8}+\sqrt{2}$

c $\sqrt{18}+\sqrt{8}$

d $2\sqrt{20}+2\sqrt{5}$

3 Simplify the following:

a $\sqrt[3]{512v}-5\sqrt[3]{v}$

b $2\sqrt[3]{u}+\sqrt[3]{125u}$

c $\sqrt{ax^5}+x^2\sqrt{ax}$

Multiplication and division of radicals

4 Simplify the following:

a $\sqrt{2}\times\sqrt{11}$

b $2\sqrt{2}\times 2$

c $4\sqrt{35}\times\sqrt{5}$

d $5\sqrt{7}\times 3\sqrt{5}$

e $\sqrt{5}\times\sqrt{2}\times\sqrt{7}$

f $\sqrt{343}\times\sqrt{75}$

g $\sqrt{5}\times\sqrt{7}$

h $8\times 10\sqrt{5}$

i $\sqrt{7}\times\sqrt{3}\times\sqrt{11}$

j $\sqrt{55}\times\sqrt{11}$

k $4\sqrt{11}\times 5$

l $2\sqrt{5}\times 15\sqrt{11}$

m $7\sqrt{22}\times\sqrt{2}$

n $\sqrt{180}\times\sqrt{48}$

o $8\sqrt{15}\times 8\sqrt{5}$

p $5\sqrt{17}\times 8\sqrt{3}$

q $17\sqrt{35}\times 4\sqrt{5}$

r $8\sqrt{51}\times 9\sqrt{3}$

5 Simplify the following:

a $\sqrt{85}\div\sqrt{17}$

b $18\sqrt{7}\div 9$

c $24\sqrt{112}\div 3\sqrt{7}$

d $8\sqrt{77}\div\sqrt{7}$

e $20\sqrt{14}\div 5\sqrt{2}$

f $\sqrt{15}\div\sqrt{5}$

g $\sqrt{55}\div\sqrt{5}$

h $\sqrt{51}\div\sqrt{17}$

i $\sqrt{21}\div\sqrt{3}$

j $\sqrt{91}\div\sqrt{7}$

k $40\sqrt{7}\div 8$

l $10\sqrt{55}\div\sqrt{11}$

m $15\sqrt{22}\div\sqrt{11}$

n $4\sqrt{35}\div 2\sqrt{5}$

o $\sqrt{27}\div\sqrt{3}$

p $3\sqrt{20}\div\sqrt{5}$

q $5\sqrt{8}\div\sqrt{2}$

r $40\sqrt{96}\div 10\sqrt{6}$

s $50\sqrt{24}\div 10\sqrt{6}$

t $\sqrt{25}\div\sqrt{81}$

6 Simplify the following:

a $\sqrt{\dfrac{28}{7}}$

b $\sqrt{\dfrac{9}{45}}$

c $\sqrt{\dfrac{64}{4}}$

d $\sqrt{\dfrac{48}{144}}$

e $\dfrac{\sqrt{12}}{\sqrt{36}}$

f $\dfrac{\sqrt{56}}{\sqrt{14}}$

g $\dfrac{\sqrt{36}}{\sqrt{81}}$

h $\dfrac{\sqrt{72}}{\sqrt{32}}$

i $\dfrac{17\sqrt{21}}{\sqrt{3}}$

7 Assuming that all variables are non-negative, simplify the following:

a $\sqrt{4j}\sqrt{j}\sqrt{4k}$

b $y\sqrt{yy}\sqrt{y}$

8 Simplify the following:

a $\dfrac{\sqrt[3]{6}}{\sqrt[3]{6 \times 7^3}}$

b $\dfrac{\sqrt{2p}}{\sqrt{2q}}$

c $6\sqrt{7m} \div \sqrt{4m}$

Mixed operations with radicals

9 Distribute and simplify the following:

a $\sqrt{7}\left(-5+\sqrt{5}\right)$

b $7\sqrt{2}\left(\sqrt{7}+6\right)$

c $7\sqrt{11}\left(-6\sqrt{2}+\sqrt{7}\right)$

d $\sqrt{11}\left(\sqrt{7}+4\right)$

e $\sqrt{7}\left(3+\sqrt{3}\right)$

f $\sqrt{2}\left(\sqrt{11}-6\right)$

g $3\sqrt{3}\left(\sqrt{13}-5\right)$

h $\sqrt{3}\left(\sqrt{11}+\sqrt{13}\right)$

i $4\sqrt{7}\left(\sqrt{2}-\sqrt{11}\right)$

j $3\sqrt{5}\left(\sqrt{55}+\sqrt{11}\right)$

k $8\sqrt{2}\left(\sqrt{3}-3\sqrt{7}\right)$

l $5\sqrt{2}\left(3\sqrt{5}+4\sqrt{7}\right)$

m $7\sqrt{3}\left(\sqrt{15}+\sqrt{60}\right)$

n $11\sqrt{3}\left(3\sqrt{5}-\sqrt{20}\right)$

o $8\sqrt{11}\left(3\sqrt{7}-4\sqrt{5}\right)$

p $\sqrt{13}\left(\sqrt{11}+2\right)$

q $8\sqrt{11}\left(-3\sqrt{5}-7\sqrt{7}\right)$

10 Distribute and simplify the following:

a $\left(\sqrt{13}+5\right)\left(\sqrt{2}-3\right)$

b $\left(\sqrt{7}-\sqrt{13}\right)\left(\sqrt{2}-\sqrt{11}\right)$

c $\left(\sqrt{11}+1\right)^2$

d $\left(2\sqrt{5}+3\right)^2$

e $\left(5-\sqrt{7}\right)\left(8-\sqrt{3}\right)$

f $\left(3\sqrt{11}-\sqrt{5}\right)^2$

g $\left(\sqrt{11}-11\right)\left(\sqrt{11}+11\right)$

h $\left(\sqrt{10}+\sqrt{7}\right)\left(\sqrt{10}-\sqrt{7}\right)$

11 Simplify the following:

a $3\sqrt{5} \times 8\sqrt{10} \div 2\sqrt{2}$

b $\dfrac{8\sqrt{12} \times 3\sqrt{2}}{4\sqrt{10}}$

c $\dfrac{\sqrt{15} + \sqrt{21}}{\sqrt{3}}$

d $\dfrac{3\sqrt{2} \times 2\sqrt{6}}{\sqrt{8} \times \sqrt{20}}$

12 Find the values of x and y in the following equations:

a $\left(8\sqrt{91} - 5\sqrt{7}\right)^2 = x + y\sqrt{13}$

b $\left(\sqrt{x} + y\right)^2 = 14 - 6\sqrt{5}$

13 Consider the expression $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$.

a Distribute and simplify the expression.

b What value of m can be substituted into $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$ so that it has a rational value?

14 Simplify the following:

a $\sqrt[3]{u^2}\sqrt[3]{u}$

b $\sqrt[3]{3u^2}\sqrt[3]{9u}$

c $7\sqrt{u} + 3\sqrt{u}$

d $10\sqrt{u} - 8\sqrt{v} - 4\sqrt{u} + 9\sqrt{v}$

e $6\sqrt{a} + \sqrt{81a}$

f $\sqrt{125a} - \sqrt{20a}$

Rationalize the denominator

15 State whether the following results in a rational number:

a $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} + \sqrt{3}\right)$

b $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} - \sqrt{3}\right)$

c $\left(-\left(\sqrt{8} + 2\right)\right) \times \sqrt{8}$

16 Rationalize the denominator of the following:

a $\dfrac{1}{\sqrt{11}}$

b $\dfrac{\sqrt{2} - 2}{\sqrt{11}}$

c $\dfrac{7}{\sqrt{12}}$

d $\dfrac{14\sqrt{11}}{9\sqrt{13}}$

e $-\dfrac{3}{5\sqrt{13}}$

f $\dfrac{7\sqrt{30}}{\sqrt{10}}$

g $\dfrac{10\sqrt{7} - 1}{\sqrt{5}}$

h $\dfrac{\sqrt{5}}{\sqrt{15}}$

17 Find a conjugate for the following expressions:

a $2 + \sqrt{7}$

b $8 - \sqrt{u}$

c $4 - 3\sqrt{10}$

d $5\sqrt{3} + 4\sqrt{u}$

18 Rationalize the denominator for the following expressions:

a $\dfrac{9}{\sqrt{5}-7}$

b $\dfrac{28}{\sqrt{10}-\sqrt{3}}$

c $\dfrac{\sqrt{7}}{5\sqrt{6}+4\sqrt{2}}$

d $\dfrac{\sqrt{11}+\sqrt{2}}{\sqrt{11}-\sqrt{2}}$

e $\dfrac{9\sqrt{7}+8\sqrt{5}}{9\sqrt{7}-8\sqrt{5}}$

f $\dfrac{8}{9\sqrt{11}-5}$

g $\dfrac{\sqrt{7}+\sqrt{2}}{\sqrt{7}-\sqrt{2}}$

h $\left(\sqrt{10}+1\right)^{-1}$

19 Express the following in simplest form with rational denominator:

a $\dfrac{3}{\sqrt{10}}+\dfrac{1}{4}$

b $\dfrac{3}{4\sqrt{5}}-\dfrac{3}{2\sqrt{2}}$

c $\dfrac{5\sqrt{3}}{6\sqrt{5}}-\dfrac{7\sqrt{5}}{3\sqrt{3}}$

d $\dfrac{1}{\sqrt{2}+5}+\dfrac{1}{\sqrt{2}-5}$

20 Show that the following expression is a rational number:

$$\frac{3}{\sqrt{2}+3}+\frac{3}{\sqrt{2}-3}$$

21 If $x=2+\sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a $\dfrac{x-\sqrt{3}}{x+\sqrt{3}}$

b $\left(x+\dfrac{1}{x}\right)^2$

22 If $x=4+\sqrt{3}$, simplify and rationalize the denominator of the following expressions:

a x^2

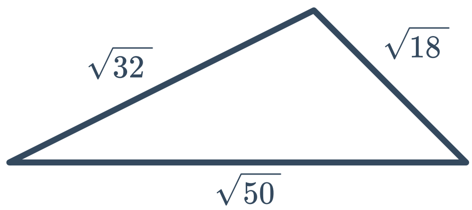
b $\dfrac{x+1}{x-1}$

23 Express the following as a single radical:

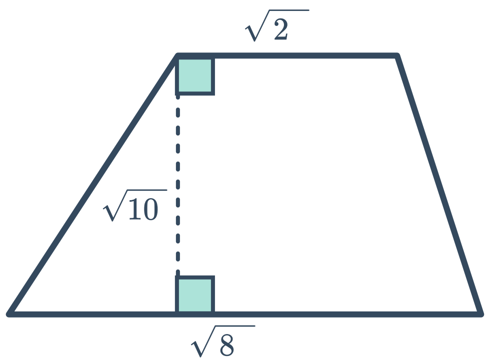
$$42+\frac{6}{7+5\sqrt{2}}$$

Applications

24 Find the perimeter of the triangle in simplified radical form:



25 Find the area of the trapezoid in radical form:

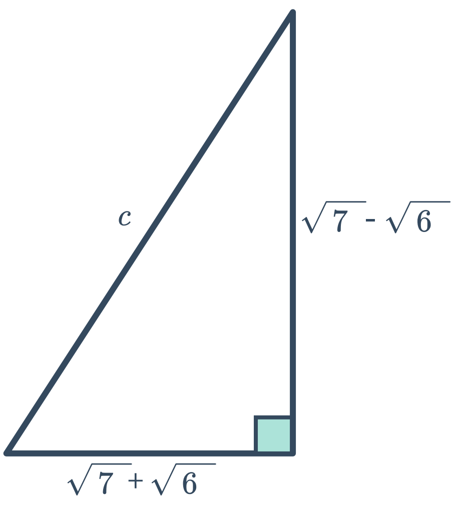


26 A rectangle has a width of $\left(\sqrt{7}-2\right)$ ft and a length of $\left(\sqrt{7}+2\right)$ ft.

a Find the exact perimeter of the rectangle.

b Find the exact area of the rectangle.

27 Find c, the length of the hypotenuse of the triangle below. Leave your answer in radical form.



28 Find the exact perpendicular height of a triangle whose area is $36\sqrt{15}$ square centimeters and whose base measures $8\sqrt{5}$ centimeters.

29 A rectangle with height h and width w is believed to be more aesthetically pleasing if its dimensions are in the following ratio:

$$\frac{w}{h} = \frac{2}{\sqrt{5} - 1}$$

a Rationalize the denominator of the ratio.

b Find a decimal approximation for this ratio correct to the nearest thousandth.

30 A rectangle has a length of $\left(\sqrt{10} + 3\right)$ meters and an area of 4 square meters. What is the exact width of this rectangle? Give your answer with a rational denominator.

31 The formula $P = \dfrac{x\left(12 + \sqrt{x}\right)}{5\sqrt{x}}$ models the percentage, P, of bees that do not survive relocation to a new hive in a colony that has x thousand bees.

a In a hive with $25\,000$ bees, what percentage will not survive relocating to a new hive? Round your answer to the nearest integer.

b Rewrite the formula for P by rationalizing the denominator.

c Using the formula you found in part (b), find the percentage of bees from a hive with $25\,000$ bees that will not survive relocation. Round your answer to the nearest integer.

32 Consider a square pyramid with a height h and a volume V. The length of one side of its base is given by $\sqrt{\dfrac{3V}{h}}$.

a Rewrite the expression for the length such that it has a rational denominator.

b Find the side length of the base of a square pyramid that has a height of 3 inches and a volume of 16 cubic inches.